

Math 403 Homework #4, Spring 2025

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Solutions by: ...your name...

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(1 point for each of up to 3 collaborators who also list you)

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Due: 11:59pm Saturday 22 March 2025

READING ASSIGNMENTS

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16–17. for Tue. 4 March and Thu. 6 March

- [Tapp, Chap.5] Lie algebras as tangent spaces to the identity
- [Lax, Chap.9: Matrix-valued Functions, through §Matrix Exponential]
- [Wikipedia, *Exponential map (Lie theory)*]

18. for Tue. 18 March

- [Lax, Chapter 16, p.237–240] entrywise positive matrices, Perron's theorem

19. for Thu. 20 March

- [Lax, Chapter 16, p.240–245] stochastic and nonnegative matrices, Frobenius thm

20. for Tue. 25 March

- [Treil, §8.5] multilinear algebra, tensor product
- [Wikipedia, *Tensor product*]

21–22. for Thu. 27 March and Tue. 1 April

- [Wikipedia, *Exterior algebra*]

EXERCISES

1. Show that $\|AB\|_2 \leq \|A\|_2\|B\|_2$ whenever the product AB is defined.

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Solution

$$\|A\|_2 = \sigma_{\max}(A)$$

Since the spectral norm is the max stretching factor of the matrix, the product can't stretch more than the product of the individual matrices. If a more rigorous proof is needed,

assume that it is false, and that there is a vector x such that $|ABx| > \|A\|_2\|B\|_2x$. But the $|Bx|$ is bounded by $\|B\|_2x$, so ABx is bounded by $\|B\|_2Ax$, and Ax itself is bounded by $\|A\|_2x$, so we get a contradiction, because $|ABx| \leq \|B\|_2\|Ax\| \leq \|A\|_2\|B\|_2x$. : A more rigorous proof is indeed needed

2. Show that $\lim_{k \rightarrow \infty} A^k = 0$ if and only if its spectral radius satisfies $\rho(A) < 1$.

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Part II not quite rigorous

Solution

$$\rho(A) = \max\{|\lambda| \mid \lambda \text{ is an eigenvalue of } A\}$$

Claim($\rightarrow$): $\lim_{k \rightarrow \infty} A^k = 0 \rightarrow \rho(A) < 1$.

Assume that A has eigen value $\lambda \geq 1$, and $x \in \text{eigenspace}(\lambda)$, then $\lim_{k \rightarrow \infty} A^k x = \lambda^k x$, which isn't equal to 0, unless x is the zero vector which can't happen because it is an eigenvector.

Claim($\leftarrow$): $\rho(A) < 1 \rightarrow \lim_{k \rightarrow \infty} A^k = 0$

The generalized eigen vectors of A provide a basis for the entire ambient space. Describe every vector as a linear combination of these generalized eigenvectors. Since $\rho(A) < 1 \rightarrow \lim_{k \rightarrow \infty} |A^k x| = 0$, for all vectors x , since the generalized eigen vectors all go to 0. Now take the 'Standard' basis and multiply them with $\lim_{k \rightarrow \infty} A^k$, they are all 0, which means the columns are 0, therefore the matrix is 0.

3. Show that if ν is a consistent norm on $\mathbb{C}^{n \times n}$, then $\lim_{k \rightarrow \infty} \nu(A^k)^{\frac{1}{k}} = \rho(A)$.

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Solution

Let $A^* = \frac{A}{\rho(A)+x}$, where $x > 0$. Since $\rho(A/k) = |1/k|\rho(A)$, $\rho(A^*) < 1$. By the results of the previous question $\lim_{k \rightarrow \infty} (\nu((A^*)^k)) < 1$, since it is submultiplicative.

This also means also $(\nu((A^*)^k))^{1/k} < 1$. Decomposing to our original expression, $(\rho(A)+x)\nu((A^*)^k)^{1/k} < 1$, if you take the sequence $x = 1/k$, we see that this converges to $\nu(A^k)^{1/k} = \rho(A)$ since $\rho(A^k) \geq \rho(A)^k$, for any consistent norm ν .

4. Prove or disprove, and salvage the statement as best you can in case you find it's false: 0/3

If $\|(\tilde{\lambda}I - A)^{-1}\| \geq \eta$ then there is an eigenvalue λ of $A \in \mathbb{C}^{n \times n}$ satisfying

$$|\tilde{\lambda} - \lambda| \leq 2(\|A\| + \eta^{-1})\eta^{-\frac{1}{n}}.$$

Solution

5. For $A \in \mathbb{C}^{m \times n}$, set $\|A\|_\infty = \max_{\|x\|_\infty=1} \|Ax\|_\infty$. Prove that $\|A\|_\infty = \max_i \sum_j |a_{ij}|$.

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Not quite

Solution

Claim #1: All the elements of x are conjugates of all the elements in a row.

Assume the largest absolute value entry happens in in the row j .

$$\begin{aligned}\|Ax\| &= \|(Ax)_j\| \\ (Ax)_j &= A_jx\end{aligned}$$

This is a sum of values where each value is bounded by the absolute values of the numbers being multiplied. If you make the entries of x the conjugates then you get the max value of this expression, and through construction we know that this is the max value that can be achieved since the absolute value is bounded by the product of the absolute values(modules), and since the infinity norm of x is 1, then the max the value could be is the absolute value of the entries in each row. Which we have found a valid construction. Each element of x is also normalized to be on the unit circle.

6. Assume that the union of m out of the n Gerschgorin disks $\mathcal{G}_i$ is disjoint from the other $n - m$ disks. Prove that the union of m disks contains precisely m of the eigenvalues. Hint: how do the eigenvalues move as A proceeds along a straight line to $\bar{A}$?

Not rigorous: you need an explicit deformation and an explicit use of continuity.

Solution

Start off with a matrix that only contains the elements on the diagonal. The elements on the diagonals are the eigenvalues and the union of all the Gerschgorin disks are just the points. Since we will take the union of m of the Gerschgorin disks, find the rows from which we will get the radius.

Slowly add multiplies of the these rows subtraced the diagonal to the matrix. So we are slowly 'moving' towards A . Since the deformation of the eigenvalues is contininuous the eigenvalues can't 'jump' out of the union of the m disks because if they did, they would have to be in another Gerschgorin disk, but since the m Gerschgorin disks are disjoint from the ones, it would be left in no Gerschgorin disk, which is a contradiction.

So once you finish adding back all the elements in the rows of the Gerschgorin disk, you can start adding the other rows subtracted the diagonal back. By the same reasoning the eigenvalues can't leave the union of the $n - m$ disks, and in the end. You get back A with m eigenvalues in the union of m of the Gerschgorin disks and the other $n - m$ in the other disks.

Finer note about 'jumping', the union of the m Gerschgorin disks is closed union of circles/spheres, and the other $n - m$ Gerschgorin disks are just points. Since the eigenvalues move contininously, it would have to first be on the boundary of the union of the m Gerschgorin disks and then move towards a point(Since the other $n-m$ Gerschgorin disks are just points). You will have a nonzero distance between these points and the boundary since they are disjoint, so the eigenvalue would have to move this distance to get to the other points. But that would imply that it would be in an area where it isn't in the union of any Gerschgorin disks, which is the contradiction.

7. Let $G(A)$ be the union of the Gerschgorin disks of A . Show that the intersection $\bigcap_S G(S^{-1}AS)$ over all nonsingular matrices S equals the spectrum of A .

Solution

First the eigenvalues will always be in the Gerschgorin disks since change of basis doesn't change the eigen values.

Assume there is some element except the eigenvalues in the intersection. Take the jordan canonical form of the matrix. Since there are eigenvalues on the diagonal, the Gerschgorin disks are centered around the eigenvalues and the radius is at max 1. Since there are only 1's in the superdiagonal. Now change the basis of so that the ones become arbitrarily small, so x can't be arbitrarily close to the eigenvalues, therefore only the eigenvalues are in the intersection. You need to explain how to do this

8. Prove that the spectrum of A is contained in $G(A) \cap G(A^T)$. Illustrate with the 3×3 matrix with entries $a_{ij} = i/j$. **Solution**

The characteristic polynomial of A and A^T are the same. So the spectrum is in the intersection.

$$A = \begin{bmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 2/3 \\ 3 & 3/2 & 1 \end{bmatrix}$$

The reduced row echelon form of the matrix A is:

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

The reduced row echelon form of the transpose of A is:

$$\text{rref}(A^T) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

So the eigen values are the same.

9. Fix the matrix

$$A = \begin{bmatrix} 7 & -16 & 8 \\ -16 & 7 & -8 \\ 8 & -8 & -5 \end{bmatrix}.$$

- (a) Use Gerschgorin's Theorem to say as much as you can about the locations of the $3/3$ eigenvalues of and the spectral radius of A .

Solution

The eigen values of A are within two circles, centered around 7 of radius 24 and -5 of radius 16.

- (b) Consider $D^{-1}AD$ for a diagonal matrix D to see if you can improve your estimates $3/3$ for the eigenvalue locations.

Solution

$$\begin{bmatrix} 99 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 201 \end{bmatrix} \begin{bmatrix} 7 & -16 & 8 \\ -16 & 7 & -8 \\ 8 & -8 & -5 \end{bmatrix} \begin{bmatrix} 1/99 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1/201 \end{bmatrix} = \begin{bmatrix} 7 & -176 & \frac{264}{67} \\ \frac{-16}{11} & 7 & \frac{-24}{67} \\ \frac{536}{33} & \frac{-536}{3} & -5 \end{bmatrix}$$

I think so? I don't get a better approximation.

- (c) Compute the actual eigenvalues and comment on the quality of your estimates $3/3$ in (a) and (b).

Solution

The eigenvalues are 27, -9, -9. I guess I should have guessed that the two of the eigen values are the same because the disks are the same. But I don't have a proof for that. I also see that one of the eigen values is very far from the center.

10. Find the Lie algebra $\mathfrak{so}_n$ of the special orthogonal group $SO_n(\mathbb{R})$.

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This is right but it doesn't give the actual Lie algebra unless you also compute the Lie algebra of $O(n)$

Solution

The Lie algebra of the orthogonal group is set of the derivatives of all paths through the origin. Since any path along the origin inside the orthogonal group has to have determinant one (The determinant is continuous and therefore there is no discontinuous jump between the matrices of determinant 1 and -1, and the determinant of the identity is 1), the Lie algebra of the special orthogonal group is just the Lie algebra of the Orthogonal group, this group is also a subset of the Lie algebra of the special linear group since the elements along the diagonal are zero in a skew matrix.

11. Fix a matrix group $G \subseteq M_n F$ over the field $F \in \{\mathbb{R}, \mathbb{C}\}$ and a matrix $A \in G$. Show $3/3$ that the tangent space $T_A(G)$ of G at A is $A\mathfrak{g}$, and show that this equals $\mathfrak{g}A$.

Solution

Claim 1: $A\mathfrak{g} \subseteq T_A(G)$.

For every element in the Lie algebra, there exists a path that crosses through the origin and has that element as a derivative. If you multiply this path by A, then this path crosses through A at 0, and is also in the group, the derivative of this new path is the old derivative times A, so $A\mathfrak{g} \subseteq T_A(G)$.

Claim 2: $T_A(G) \subseteq A\mathfrak{g}$

Take a path in $T_A G$. At 0 the path is A. Since A is in a group it must have an inverse, multiply this path by the inverse, and then you get path that crosses the origin at 0, and this path is in $\mathfrak{g}$. So when you multiply it by A, you get back to the previous path, proving the lemma.

This proof also works for $\mathfrak{g}A$, so they are equal (since $Ag = T_A(G) \wedge gA = T_A(G) \rightarrow gA = Ag$).

12. Prove that $\det(e^A) = e^{\text{tr}(A)}$.

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Solution

Take the jordan block of A. So $A = P^{-1}JP$.

$$\begin{aligned} \det(e^A) &= \det\left(\sum_{k=0}^{\infty} \frac{A^k}{k!}\right) = \det\left(P^{-1} \sum_{k=0}^{\infty} \frac{J^k}{k!} P\right) \\ &= \det(P) * \det\left(\sum_{k=0}^{\infty} \frac{J^k}{k!}\right) * \det(P^{-1}) \end{aligned}$$

Since J, is upper diagonal $\det\left(\sum_{k=0}^{\infty} \frac{J^k}{k!}\right) = e^{\lambda_1 + \lambda_2 + \dots + \lambda_n}$ (J upper diagonal $\rightarrow$ the elements in the diagonal become $e^{\lambda_1}, e^{\lambda_2}, \dots$ and there are no elements in the lower diagonal), which is just $e^{\text{tr}(A)}$

13. Explain why $\mathfrak{su}_n = \mathfrak{u}_n \cap \mathfrak{sl}_n(\mathbb{C})$. (You may use the previous exercise.)

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Solution

Claim 1: $\mathfrak{su}_n \subseteq \mathfrak{u}_n \cap \mathfrak{sl}_n$

This is trivially true by definition of the SU .

Claim 2: $\mathfrak{u}_n \cap \mathfrak{sl}_n \subseteq \mathfrak{su}_n$

Take $A \in \mathfrak{u}_n \cap \mathfrak{sl}_n$.

Take the path realization of A. This path is in both U_n and SL_n (By lecture 16 and lecture 17) and therefore is also in SU_n . So e^{tA} is both in U_n and SL_n .

You've gotta say a little more

14. Let $\gamma : (-\varepsilon, \varepsilon) \rightarrow GL_n(F)$ be a differentiable path, where $F \in \{\mathbb{R}, \mathbb{C}\}$. Use Cramer's [3/3](#) rule (or some other method) to show that the inverse path $t \mapsto \gamma(t)^{-1}$ is differentiable.

Solution

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Since the determinant is never zero and the cofactor and determinant are a rational function. The derivative is always defined.